"Heaven's Light is Our Guide"

OBE Evaluation and Grading Webapp

ECE-2100: Software Development Project-I

Submitted By

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Introduction to OBE

Outcome-Based Education

A student-centered learning model that focuses on what students can **demonstrate** after completing a program.

- Required by the **Washington Accord** and **BAETE** for engineering accreditation in Bangladesh.
- Measures achievement through **Course Outcomes (CO)** and **Program Outcomes (PO)**.
- Enables data-driven **Continuous Quality Improvement (CQI)**.
- Shifts focus from *what is taught* to *what students can do*.

The Problem Statement

Current Challenges:

- Manual, error-prone Excel spreadsheets
- Data scattered across files and folders
- Time-consuming administrative burden
- No real-time visibility for students

Impact:

- Calculation mistakes affect accreditation
- Faculty burn hours on repetitive tasks
- Students graduate without knowing their PO gaps
- No systematic feedback loop for CQI

Project Objectives

1. **Automate** complex CO–PO attainment calculations with an algorithmic engine.
2. **Centralize** all academic data — enrollments, marks, attendance, feedback.
3. **Provide** role-specific, intuitive dashboards for 4 user types.
4. **Enable** real-time progress tracking for students.
5. **Facilitate** Continuous Quality Improvement through bidirectional analytics.

Our Solution

The OBE Evaluation & Grading Webapp

A comprehensive, role-based web platform that automates the **entire OBE lifecycle** — from course creation to final accreditation reporting.

- **Four User Panels:** Central Admin, Department Admin, Teacher, Student
- **Automated Engine:** Real-time grade and attainment processing
- **Cloud Deployed:** Accessible from any device, anywhere
- **PWA Ready:** Installable on mobile devices

Technology Stack — MERN

Frontend

- React 19 + Vite
- TailwindCSS 4
- React Router 7
- Recharts (analytics)

Authentication

- Firebase Auth
- JWT tokens

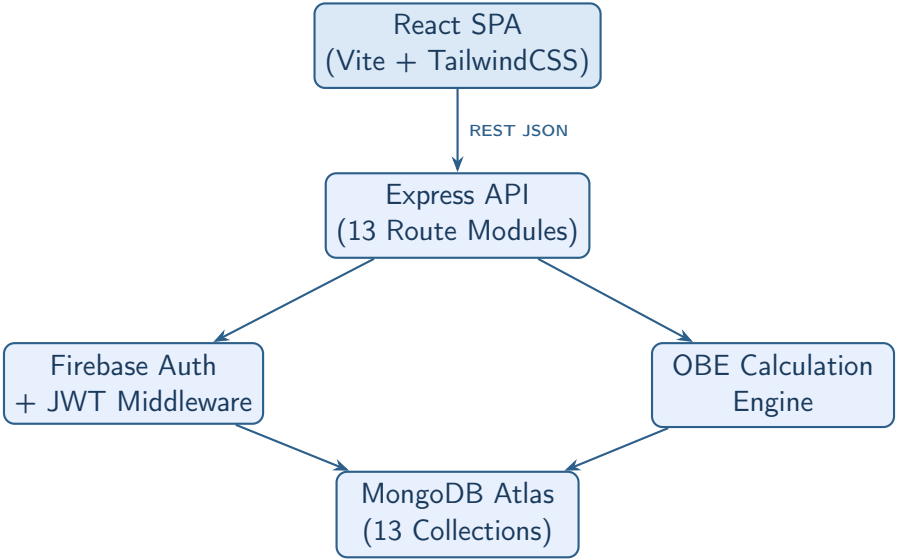
Backend

- Node.js + Express 5
- 13 RESTful route modules
- Helmet, CORS, rate-limiting

Database

- MongoDB Atlas
- Mongoose ODM
- 13 interconnected collections

System Architecture



Role-Based Access Control (RBAC)

Role	Capabilities
Central Admin	Global data management: departments, series, teachers, students
Dept Admin	Departmental operations: course assignments, feedback analytics
Teacher	Course delivery: attendance, marks, CO mappings, reports
Student	Personal tracking: grades, PO attainment, anonymous feedback

Security Model

Every API request passes through JWT authentication $\rightarrow$ role verification middleware. Routes are strictly isolated per role.

Administrative Workflows

Central Admin:

- KPI Dashboard: total departments, courses, teachers, students
- Bulk student enrollment via spreadsheet-style grid
- Series and semester management

Department Admin:

- Assign teachers to theory and sessional courses
- View Feedback Analytics with gap analysis
- Manage course advisors and publish notices

Teacher Workflow — Course Setup

1. **Define Course Outcomes (COs):** Add descriptions and map each CO to relevant POs.
2. **Create Assessments:** Set up Class Tests, Quizzes, Assignments, Finals.
3. **Map to COs:** Link each assessment to specific Course Outcomes.
4. **Configure Marks:** Define total marks and component weights.

Key Feature

The system remembers all CO $\rightarrow$ PO mappings, so the final OBE calculations are fully automated once marks are entered.

Teacher Workflow — Data Entry

Spreadsheet-Style Interface:

- Attendance: Toggle P/A per student per date, auto-computes percentages
- Marks: Enter scores in a familiar grid layout, type 'A' for absent
- One-click save with asynchronous persistence

Smart Automation:

- **Best-of-N Logic:** Automatically selects the best 3 out of 4 class tests
- **Auto-Grading:** Computes letter grades from total marks
- **Export:** Download evaluation reports as Excel or PDF

The OBE Calculation Engine

Core Pipeline

Raw Marks $\longrightarrow$ CO Attainment $\longrightarrow$ PO Attainment

Step-by-Step:

1. Collect marks from all assessments mapped to each CO.
2. Compute weighted average per CO per student.
3. Compare against faculty-defined attainment threshold.
4. Aggregate CO attainment into PO attainment percentages.
5. Generate the final **Evaluation Report** with grades.

Eliminates hours of manual calculation entirely.

Student Experience

Personalized Dashboard:

- View all enrolled courses with attendance percentages
- Aggregate PO attainment bars across completed courses
- Per-course breakdown: marks so far, CO attainment status

Transparency & Engagement:

- Students see *exactly* which outcomes they achieved or missed
- Access attendance logs, marksheets, and OBE attainment data
- Submit anonymous feedback on course quality

Continuous Quality Improvement (CQI)

Student Feedback

- 6 standardized Likert-scale questions
- Fully anonymous submissions
- Published by teacher, controlled window

Instructor Reports

- Self-rating across 4 categories
- Written analysis per category
- Submitted to department admin

Gap Analysis

Department Admins view a side-by-side comparison of student satisfaction vs. instructor self-rating, identifying courses that need intervention.

Security & Performance

Security

- Firebase Auth + JWT tokens
- Helmet.js HTTP headers
- Rate limiting on all endpoints
- Input sanitization (XSS, NoSQL injection)
- CORS whitelist

Performance

- MongoDB indexing on hot queries
- Optimized aggregation pipelines
- React lazy-loading & code splitting
- PWA: installable on mobile
- Netlify CDN edge delivery

Development Methodology

Agile Approach

- 6-week iterative development
- Feature-driven milestones
- Frequent peer code reviews
- Continuous deployment to Netlify

Team of 6

- 2 Backend developers
- 2 Frontend developers
- 1 Integration developer
- 1 UI/UX & QA developer

Version Control

Git/GitHub with Conventional Commits, feature branches, and pull request reviews.

Future Scope

- **Automated BAETE Reports:** One-click PDF generation for accreditation submissions.
- **Mobile Application:** Dedicated React Native apps for iOS and Android.
- **AI-Powered Analytics:** Predictive modeling to identify at-risk students early.
- **University Integration:** Syncing with RUET's central automation system.
- **Multi-University Support:** Extend the platform beyond RUET to other institutions.

Conclusion

What We Built

A modern, scalable web platform that **replaces manual spreadsheets** with an automated OBE engine, serving all stakeholders from administration to students.

Key Takeaways:

- Ensures RUET's compliance with global accreditation standards
- Saves hundreds of administrative hours per semester
- Empowers students with real-time outcome transparency
- Closes the feedback loop for continuous improvement

Thank You!

Questions & Live Demonstration

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